

Last modified on

# Sports Data Soccer API - Match Tracking Feed (MA25)

Get tracking data giving the coordinates of the players, referee, and ball position during a given match.

## Overview

The Soccer **Match Tracking feed** provides X/Y coordinate data showing the position of all players and the referee on the pitch at a given point in time, and also provides the XYZ coordinates of the ball position. Files will be provided and available to download after the match has ended (post match).

The two files that will be available are:

- 10 frames per second (FPS)
- 25 frames per second (FPS)

The URLs will be shown in the feed output and will look similar in appearance to the below example snippets. You must copy and paste the URL into a web browser (into its address bar) to initiate the download.

### XML:

```
<matchTracking>
  <matchInfo id="7ijuqohwgmplbxjdj1625sxwfe" ...>
    ...(sample is intentionally truncated)...
  </matchInfo>
  <trackingFiles>
    <fps10>http://api.performfeeds.com/soccerdata/matchtracking/download/soccer_10fps_7ijuqohwgmp
    plbxjdj1625sxwfe.txt?token=st=1605706968~exp=1605707568~acl=/*~hmac=78720ba1a7702628367c7a81e204a55b9
    a50ac19c4b3e68929c08af6abc16754</fps10>
    <fps25>http://api.performfeeds.com/soccerdata/matchtracking/download/soccer_25fps_7ijuqohwgmp
    plbxjdj1625sxwfe.txt?token=st=1605706969~exp=1605707569~acl=/*~hmac=d6387f70313b36a1358cb60e8a20b06e8
    245cbab5ed5a432f0f52d372d954a59</fps25>
  </trackingFiles>
</matchTracking>
```

### JSON:

```
{
  "matchInfo": {
    "id": "djg1nlbnxsept5d880l2nszc9",
    ...(sample is intentionally truncated)...
  },
  "trackingFiles": {
    "fps10": "http://api.performfeeds.com/soccerdata/matchtracking/download/soccer_10fps_7ijuqohwgmp1bxdj1625sxwfe.txt?token=st=1605706968~exp=1605707568~acl=/*~hmac=78720ba1a7702628367c7a81e204a55b9a50ac19c4b3e68929c08af6abc16754",
    "fps25": "http://api.performfeeds.com/soccerdata/matchtracking/download/soccer_25fps_7ijuqohwgmp1bxdj1625sxwfe.txt?token=st=1605706969~exp=1605707569~acl=/*~hmac=d6387f70313b36a1358cb60e8a20b06e8245cbab5ed5a432f0f52d372d954a59"
  }
}
```

## What Does The Data In The Tracking Files Represent?

Each tracking file is a comma-separated value (CSV) file with a `.txt` file extension. These are large files - which is one reason that the delivery method is via a file download. When you open a file, you will see that its contents comprises many rows and columns of characters.

To get a breakdown of the file structure and what the data represents, see: [Understanding the tracking file data structure](#)

Example: `.txt` file excerpt, rows 1 to 3:

```
1599853500000;0,1,0:0,348698,21,37.962,50.809;0,459425,6,36.156,26.631;0,537672,5,34.75,37.597;0,556911,26,41.612,33.251;0,601891,10,44.582,40.71;0,605190,8,47.719,25.374;0,623384,77,51.091,41.901;0,836014,9,51.18,23.7;0,914225,23,37.778,14.238;0,1161173,34,51.222,13.632;1,376690,24,61.252,30.377;1,620058,36,76.996,26.419;1,732829,34,51.415,32.433;1,794952,11,51.915,6.252;1,819087,25,66.154,60.634;1,892081,30,57.562,39.632;1,915206,14,64.076,9.673;1,924246,5,76.08,43.542;1,952426,6,65.737,33.636;1,1058446,10,50.795,55.441;3,975799,1,3.009,31.815;4,565463,33,96.231,31.918;:51.46,32.110;
1599853500100;100,1,0:0,348698,21,37.962,50.81;0,459425,6,36.116,26.615;...(sample is intentionally truncated)...
1599853500200;200,1,0:0,348698,21,37.96,50.811;0,459425,6,36.063,26.608;...(sample is intentionally truncated)...
```

### Required Compliance:

To comply with our API security and best practice logic for your API calls please refer to our [FAQs](#). We also recommend that you take a look at our [API Overview and Authentication Guide](#). You will learn the basics of how our API works and key concepts to get started.

## How Often Should I Poll The Feed?

This is a post-match feed, so the data is provided in downloadable files; therefore, you should only call the feed only as often as you need to. You will be provided with a URL. You must copy and paste the URL from the feed output/response into a web browser (into its address bar) to initiate the file download.

| UPDATED                                                     | POLLING FREQUENCY                                                                             |            |        |            |             |          |        |            |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------|--------|------------|-------------|----------|--------|------------|
|                                                             | TIER 15                                                                                       | TIER 12+14 | TIER 4 | TIER 11+13 | TIER 8+9+10 | TIER 6+7 | TIER 5 | TIER 1+2+3 |
| After the end of a match, ensuring all information is added | Once after the end of each match (after full time) and up to 48 hours thereafter (post match) |            |        |            |             |          |        |            |

**Note:** This feed will launch with limited availability. Please ask your Account Manager for information about competition availability and about commercial legalities around this feed.

## Match Tracking (MA25) Guide

| USAGE                                                                                                         | FEED              |
|---------------------------------------------------------------------------------------------------------------|-------------------|
| GET match tracking data showing the pitch coordinates of the players, referee and ball during the given match | /soccerdata/match |

If your request URL is as follows - that is, it does not contain a fixture UUID - the feed returns an error:  
/soccerdata/matchtracking/{outletAuthKey}?\_fmt={xml|json}&\_rt={b|c}

### Query Parameters And Example Request Parameters

This feed also makes use of the [Global query parameters](#), and you can combine them with the following query parameters (some of which can be used in conjunction with each other). However, you must query the `/matchtracking` feed by fixture UUID.

The ability to query by a legacy Opta ID is not provided in the Match Tracking feed.

#### How to use query parameters and make successful requests:

In our guides, we use HTTPS syntax for our example GET requests (rather than cURL, for example). All URLs are case-sensitive - this means that URL strings must contain the correct case and operators. You MUST format your requests correctly and include certain information; otherwise, an error will be returned. You can find information about the API, including the base URL ( `{protocol}://{requestDomain}/{feedResource}` ), in the [main guide](#) and [API Overview and Authorisation guide](#).

- Include the following information in the URL or header and make sure it is correct for your outlet: your unique and valid `{outletAuthKey}` this must be after the Outlet Authentication Key (and endpoint, if applicable). for `_rt={mode}` (operating mode) and `_fmt={dataFormat}` (response format), entity query parameter and UUID.
- Begin the query parameter string with `?` (question mark) - this must be after the Outlet Authentication Key (and endpoint, if applicable). To build up a query string, make sure that each query parameter is separated by the `&` (ampersand) operator. Values must be separated by the correct operator, for example `&` (multiple values, 'AND'), or - if a parameter supports it - `,` (multiple values, 'OR') or `!` ('NOT').
- Use the required `_` (underscore) for any parameters listed with this prefix.
- Remove any placeholder value curly braces `{ }` from your calls (we include these as an example only).

- If using the JSONP format ( `_fmt=jsonp` ) you MUST use it in combination with the callback parameter ( `_clbk` ) and a fixed value (or where you use a different value in a different instance, you must do this as little as possible). Be aware that common JavaScript frameworks, such as jQuery, can default to use randomly-generated function names - you must make sure that these are overridden and define a fixed function name instead.

| Query parameter                            | Value and description                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>{fixtureUuid}</code><br>Fixture UUID | <p>GET <b>match tracking</b> data by fixture UUID (path parameter method). Pass the fixture UUID in the request URL directly as the <code>{assetId}</code> (asset/resource ID).</p> <pre>GET https://api.performfeeds.com/soccerdata/matchtracking/{outletAuthKey}/{fixtureUuid}?_rt={mode}&amp;_fmt={dataFormat}</pre> |

## Sample Calls

These sample calls let you see an example URL and the response - click on the links below (hyperlinks open in a new window).

- [XML](#)
- [JSON](#)

## Feed Output

Here, you can find an explanation of all possible response fields and data. This is XML but fields in the feed response are similar for JSON.

▶<matchTracking> (XML only)

▶<matchInfo>

▶<description>

▶<sport>

▶<ruleset>

▶<competition>

▶<country>

▶<tournamentCalendar>

▶<stage>

▶<series>

▶<contestants> (XML only)

▶<contestant>

▶&lt;country&gt;

▶&lt;venue&gt;

▶&lt;trackingFiles&gt;

▶&lt;fps10&gt;

▶&lt;fps25&gt;

## Understanding The Tracking File Data

Each tracking file is a comma-separated value (CSV) file with a `.txt` file extension. The structure of the `.txt` files is in rows, then columns. When you download and open a file, you will see that it comprises many rows and columns of characters. For example, there could be 146,853 rows in the file.

The data held in columns 1-4 comprises a timestamp, elapsed time, match period, and match status. Column 5 is an array of all the players and referees on the pitch within the given timeframe, plus their X/Y coordinates and information to help you identify each person. Column 6 contains ball position information represented by X/Y/Z coordinates.

To help you understand what the raw data represents, here's a breakdown of the file structure.

### Quick Links

- [Column 1: Timestamp](#)
- [Column 2: Elapsed time \(frameCount\)](#)
- [Column 3: Match period](#)
- [Column 4: Match status - paused/not paused](#)
- [Column 5: Players and Referees](#)
- [Column 5 - nested data: Players](#)
- [Column 5 - nested data: Referees](#)
- [Column 6: Ball position](#)
- [Understanding the tracking file data: X, Y, Z coordinates](#)

Example: `.txt` file excerpt, rows 1 to 146,853 - all columns (6):

```
1599853500000;0,1,0:0,348698,21,37.962,50.809;0,459425,6,36.156,26.631;0,537672,5,34.75,37.597;0,556911,2
6,41.612,33.251;0,601891,10,44.582,40.71;0,605190,8,47.719,25.374;0,623384,77,51.091,41.901;0,836014,9,5
1.18,23.7;0,914225,23,37.778,14.238;0,1161173,34,51.222,13.632;1,376690,24,61.252,30.377;1,620058,36,76.9
96,26.419;1,732829,34,51.415,32.433;1,794952,11,51.915,6.252;1,819087,25,66.154,60.634;1,892081,30,57.56
2,39.632;1,915206,14,64.076,9.673;1,924246,5,76.08,43.542;1,952426,6,65.737,33.636;1,1058446,10,50.795,5
5.441;3,975799,1,3.009,31.815;4,565463,33,96.231,31.918;:51.46,32.110;
1599853500100;40,1,0:0,348698,21,37.962,50.81;0,459425,6,36.116,26.615;...(sample is intentionally truncated)...
1599853500200;200,1,0:0,348698,21,37.96,50.811;0,459425,6,36.063,26.608;...(sample is intentionally truncated)...
...(sample is intentionally truncated)...
```

## Understanding The Tracking File Data

Example - columns 1-4, rows 1 to 146,853

```
1559606400000;0,1,0
1559606400040;40,1,0
1559606400080;80,1,0
...(sample is intentionally truncated)...
1559612995040;2995040,2,0
```

### Column 1: Timestamp

Each row contains a timestamp, which increments in the number of seconds at which we capture the data. The timeframe is in UNIX Epoch time and is the first entry in each row (col 1).

The file excerpt in the [overview](#) contains timestamps that increment in .100 seconds. However, for matches where the rate at which we capture the data is 25Hz, the timestamps increment in .040 seconds (or 40 ms = 1/25 seconds).

Example - column 1: rows 1 to 146,853:

| ROW                 | COLUMN 1 - TIMESTAMP (EXAMPLE) |
|---------------------|--------------------------------|
| row 1, col 1:       | 1559606400000                  |
| row 2, col 1:       | 1559606400040                  |
| row 3, col 1:       | 1559606400080                  |
| row 146,853, col 1: | 1559612995040                  |

### Column 2: Elapsed Time (FrameCount)

The second column in each row represents the elapsed time or "frameCount".

Example - column 2: rows 1 to 146,853:

| ROW                 | COLUMN 2 - ELAPSED TIME (EXAMPLE) |
|---------------------|-----------------------------------|
| row 1, col 2:       | 0                                 |
| row 2, col 2:       | 0                                 |
| row 3, col 2:       | 0                                 |
| row 146,853, col 2: | 2995040                           |

### Column 3: Match Period

The third column in each row represents the match period:

Example - column 3: rows 1 to 146,853:

| ROW                 | COLUMN 3 - MATCH PERIOD (EXAMPLE) |
|---------------------|-----------------------------------|
| row 1, col 3:       | 1                                 |
| row 2, col 3:       | 1                                 |
| row 3, col 3:       | 1                                 |
| row 146,853, col 3: | 2                                 |

Column 4: Match Status - Paused/Not Paused

The fourth column in each row indicates whether the match is paused or not - the value is Boolean, where 0 is not paused (no) and 1 is paused (yes).

Example - column 4: rows 1 to 146,853:

| ROW                 | COLUMN 4 - PAUSED/NOT PAUSED (EXAMPLE) |
|---------------------|----------------------------------------|
| row 1, col 4:       | 0                                      |
| row 2, col 4:       | 0                                      |
| row 3, col 4:       | 0                                      |
| row 146,853, col 4: | 0                                      |

Column 5: Players And Referees

Within the colon : delimiters is a 2-dimensional array, which comprises all of the players, referees on the pitch - and the ball position - within the given time frame, separated by semi-colons ( ; ).

Each row in the nested player/referee array contains nested columns, which holds data that represents person information, such as the player type, player ID, shirt number, x position, and y position. At the end of each row after the second colon : delimiter is the ball position data, given in x, y, z coordinates.

**Note:** The last row of the data is provided **post-match**. As you can see, some of the players have exited the pitch.

Example - column 5: rows 1 to 2:

| ROW           | COLUMN 5 - PLAYER, BALL, REFEREE ARRAY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| row 1, col 5: | <div>0,c975zr70gwg4jc3691iv1dsyx,15,538.421,320.6;0,efzok<br/>rggje7s9wc6hg75yc6ax,3,540.975,355.897;0,blth4klpu88<br/>kgh73abpjlgusp,4,543.71,340.392;0,b0wb7gf04synx1kq7o<br/>8gvzkoa,18,533.909,344.659;0,9mj4hy93jtxr65ldnhj684k<br/>tm,6,537.749,338.504;0,9ydkbbygq6uocors9kcdnzbp,10,<br/>525.031,313.112;0,6u2ob6fv950r1qve8uejkk2uh,9,524.8,<br/>340.304;0,3sfi8ncpruq7hnffj4p5r2qcp,11,525.059,369.3<br/>46;0,rktyecx75c47pgwy4vm3i7d,8,525.708,337.998;0,4x<br/>hnmoqgxoeel005ek6ckfs9,5,544.638,343.972;1,agw4ksz6<br/>2ddb0q7h7lon4na0a,13,509.782,334.593;1,7970fp1tpiv5e<br/>i6ic6r6vnxwq,17,525.113,356.605;1,b0me6l44boibnwpbta<br/>sxoy02x,2,507.675,362.414;1,5yt3hvbzpaqv8lrjgt4ahb78<br/>q,20,520.675,330.574;1,9za3fwnri7qk1dutb4iwmh6g9,3,5<br/>07.207,319.9;1,e5xxqwz0sdlwkkmp8bgs1k6w9,11,522.699,<br/>316.649;1,enpah2m81un1f8owntavxz8fe,9,524.78,348.87;<br/>1,czu9f4sx15hpe05isdexmiy2,16,514.399,339.685;1,5nsa<br/>g404623qznyizl8yuwkq,5,506.927,344.576;1,52tr565cd0<br/>o78pb0wy5oqphwp,10,516.321,340.849;2,52tr565cd0o78pb<br/>0wy5oqphwp,92,544.913,375;2,a74dhjtlr37f9crjw5ajpo1x<br/>x,91,506.506,305;2,c9ok0vktgtbm8ee30u7v2v2t,90,525.<br/>8,350.527;3,f0e5posm9z0oma2t3u9k1qy95,1,567.572,337.<br/>812;4,ab8jzmb5h01h7mt66h3zgm5ex,1,474.054,336.44;</div> |

| ROW           | COLUMN 5 - PLAYER, BALL, REFEREE ARRAY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| row 2, col 5: | <div>0,4xhnmoqgxoeel005ek6ckfs9,5,544.628,343.982;0,efzo<br/>krggje7s9wc6hg75yc6ax,3,540.945,355.988;0,6u2ob6fv95<br/>0r1qve8uejkq2uh,9,524.801,340.314;0,9ydkbbyggq6uocors<br/>9kcdnzbop,10,524.991,313.171;0,b1th4klpu88kgh73abpj1<br/>gusp,4,543.721,340.373;0,b0wb7gf04synx1kq7o8gvzkoa,1<br/>8,533.88,344.68;0,3sfi8ncpruq7hnffj4p5r2qcp,11,525.0<br/>61,369.378;0,rktyecx75c47pgwy4vm3i7d,8,525.658,337.<br/>987;0,9mj4hy93jtxr65ldnhj684ktm,6,537.749,338.494;0,<br/>c975zr70gwg4jc3691iv1dsyx,15,538.434,320.652;1,b0me6<br/>l44boibnwpbtasxoy02x,2,507.705,362.404;1,e5xxqwz0sd1<br/>wkkmp8bgs1k6w9,11,522.659,316.639;1,czu9f4sx15hpe05i<br/>sdexmiy2,16,514.429,339.645;1,5nsag404623qznyyizl8yu<br/>wkq,5,506.917,344.586;1,5yt3hbxvpaqv8lrjgt4ahb78q,2<br/>0,520.726,330.688;1,9za3fwnri7qk1dutb4iwmh6g9,3,507.<br/>186,319.909;1,52tr565cd0o78pb0wy5oqphwp,10,516.369,3<br/>40.94;1,7970fp1tpiv5ei6ic6r6vnxwq,17,525.065,356.62<br/>5;1,enpah2m81un1f8owntavxz8fe,9,524.824,348.829;1,ag<br/>w4ksz62ddb0q7h7lon4na0a,13,509.765,334.614;2,c9ok0vk<br/>tgtbm8ee30u7v2v2t,90,525.71,350.529;2,a74dhjtlr37f9c<br/>rjw5ajpo1xx,91,506.466,305;2,52tr565cd0o78pb0wy5oqph<br/>wp,92,544.845,375;3,f0e5posm9z0oma2t3u9k1qy95,1,567.<br/>499,337.935;4,ab8jzmb5h0lh7mt66h3zgm5ex,1,474.02,33<br/>6.385;</div> |

Column 5 - Nested Data: Players

In the example excerpt (for row 1, column 5), you can see that the fifth column (5) contains a further 5 nested columns. Here's a further breakdown of the data nested within it.

Example - Row 1, Col 5 [row 1, cols 1-5]: 0,c975zr70gwg4jc3691iv1dsyx,15,538.421,320.6;

| NESTED COLUMN                                   | NESTED DATA EXAMPLE       | DESCRIPTION                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nested column 1 (Row 1, Col 5, [row 1, col 1]): | 0                         | Object type - for example, the player type or referee: <ul style="list-style-type: none"><li>0 = home field player</li><li>1 = away field player</li><li>3 = home goalkeeper</li><li>4 = away goalkeeper</li></ul> Note: Object type 2 represents a referee - see <a href="#">Column 5 - nested data: Referees</a> |
| Nested column 2 (Row 1, Col 5, [row 1, col 2]): | c975zr70gwg4jc3691iv1dsyx | Player UUID - this is the unique ID assigned to the person in the SDAPI                                                                                                                                                                                                                                            |
| Nested column 3 (Row 1, Col 5, [row 1, col 3]): | 15                        | Shirt number/Uniform number                                                                                                                                                                                                                                                                                        |
| Nested column 4 (Row 1, Col 5, [row 1, col 4]): | 538.421                   | X-coordinate for the position of the player - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a>                                                                                                                                                                                        |



| NESTED COLUMN                                   | NESTED DATA EXAMPLE | DESCRIPTION                                                                                                                 |
|-------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Nested column 5 (Row 1, Col 5, [row 1, col 5]): | 320.6               | Y-coordinate for the position of the player - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a> |

## Column 5 - Nested Data: Referees

Referee data will also be in nested columns 1-5 of Column 5, but it follows player data. Referees can be identified by group via object type = 2 and can also be assigned uniform/shirt numbers.

In the above example for Row 1 in "Column 5 - player, ball, referee array", the referees are given in rows 21-23 (the semi-colon ; separates the rows/referees). Here's a breakdown of the data.

Example - Row 1, Col 5 [rows 21-23, cols 1-5]:

```
2,52tr565cd0o78pb0wy5oqphwp,92,544.913,375;2,a74dhjt1r37f9crjw5ajpo1xx,91,506.506,305;2,c9ok0vktgtbm8ee30u7v2v2t,90,525.8,350.527;
```

| NESTED COLUMN                                        | NESTED DATA EXAMPLE         | DESCRIPTION                                                                                                                                              |
|------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nested column 1 (Row 1, Col 5, [rows 21-23, col 1]): | 2<br>2<br>2                 | Object type: 2 = referee<br>Note: Object types 0, 1, 3, 4 represent a player - see <a href="#">Column 5 - nested data: Players</a>                       |
| Nested column 2 (Row 1, Col 5, [rows 21-23, col 2]): | 52tr565cd0o78pb0wy5oqphwp   | Referee UUID - this is the unique ID assigned to the person in the SDAPI                                                                                 |
| Nested column 3 (Row 1, Col 5, [rows 21-23, col 3]): | 15                          | Shirt number/Uniform number - these are pre-defined, as follows:<br>90 = main referee<br>91 = assistant referee (left)<br>92 = assistant referee (right) |
| Nested column 4 (Row 1, Col 5, [rows 21-23, col 4]): | 544.913<br>506.506<br>525.8 | X-coordinate for the position of the referee - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a>                             |
| Nested column 5 (Row 1, Col 5, [row 1, col 5]):      | 375<br>305<br>350.527       | Y-coordinate for the position of the referee - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a>                             |

## Column 6: Ball Position

At the end of each row after the second colon : delimiter is the ball position data (at the given timestamp), which is given in x, y, z coordinates. Here's a further breakdown of the data nested within it.

Example - Row 9, Col 6: 525,340,0;

| EXAMPLE VALUE | DESCRIPTION                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| 525           | X-coordinate of the ball - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a>                |
| 340           | Y-coordinate of the ball - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a>                |
| 0             | Z-coordinate for the height of the ball - see <a href="#">Understanding the tracking file data: X, Y, Z coordinates</a> |

## Understanding The Tracking File Data: X/Y And XYZ Coordinates

In the tracking files associated with our Match Tracking feed, positional information is expressed in X/Y coordinates for players and referees, and in XYZ coordinates for the ball position. The coordinate values comply with the Cartesian coordinate system - you can find further information about this system below.

We use a different way of expressing the position of player and the ball in terms of coordinates/pitch coordinates in some SDAPI feeds; therefore, you should not directly compare the coordinate values given in those feeds with those given in the tracking files delivered via the Match Tracking feed.

### Cartesian Coordinate System

State of the tracking data as of 24-Oct 2019. Cartesian coordinate origins that we support:

1. origin top left; y-inverted Quadrant IV: y is 0, 1, ...n from top down (SportVu raw data input)
2. origin center; Quadrants I-IV (Tracab and Second Spectrum raw data input); converted to SportVu coordinates, stored in smoothed tracking tables
3. origin center; Quadrants I-IV (TGV raw data input): y is 0, -1, ...-n from center up; y is 0, 1, ...n from center down; converted to SportVu coordinates and stored in smoothed tracking tables  
TGV y coordinate - Visual proof of how the TGV coordinate system is y-inverted
4. origin bottom left; Quadrant I (FIFA file output); converted to invert the stored Y-coordinates for output file only

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